

Powers of Ten and Place-Value Relationships

Digit-value relationships, powers of ten in exponent form, shifting digits by 10, 100 and 1000, and rounding to a place to estimate

The one idea behind every problem. Each place in our number system is worth **ten times** the place to its right, and **one tenth** of the place to its left. Everything below is that single fact used four ways.

hundred thousands	ten thousands	thousands	hundreds	tens	ones	tenths	hundredths
10^5	10^4	10^3	10^2	10^1	10^0	$\frac{1}{10}$	$\frac{1}{100}$

Reference chart. No values are filled in — use the numbers given in each problem.

Directions.

- Show the place-value thinking, not just the answer — name the places you compared or the number of places the digits moved.
- A power of ten tells you *how many places*: 10^2 means two places, 10^3 means three places.
- Write your final answer on the answer line.

1. The number 4400 is written with two 4s.

The 4 in the thousands place is worth 4000. The 4 in the hundreds place is worth 400.

How many times as much is the thousands 4 worth as the hundreds 4? Write the division you used, then give the answer.

Answer: _____

- 2.** An exponent on 10 counts how many times 10 is used as a factor.
Write 10^3 as a repeated multiplication, then give its value in standard form.

Answer: _____

- 3.** Find 47×100 .

Explain the move in place-value words: each digit of 47 shifts how many places, and in which direction? Fill in a copy of the reference chart if it helps.

Answer: _____

- 4.** The number 0.88 is written with two 8s.

The 8 in the tenths place is worth 0.8. The 8 in the hundredths place is worth 0.08.

The hundredths 8 is worth what fraction of the tenths 8? Write the division $0.08 \div 0.8$ as a decimal answer, and name the fraction it equals in words.

Answer: _____

5. Estimate $4780 + 3260$ by rounding each number to the nearest thousand — that is, to the nearest 10^3 .

Write both rounded numbers, then the estimated sum.

Answer: _____

6. Every whole number that ends in zeros can be written as a digit times a power of ten.

(a) Fill in the exponent: $60000 = 6 \times 10^{\square}$

(b) Multiply your $6 \times 10^{\square}$ back out to check it, and write that product on the answer line.

Answer: _____

7. Find 3.6×1000 .

The digits of 3.6 shift three places to the left. Show where the 3 and the 6 land, and what fills the places behind them.

Answer: _____

8. The number 55500 is written with three 5s.

The 5 in the ten-thousands place is worth 50000. The 5 in the hundreds place is worth 500. How many times as much is the ten-thousands 5 worth as the hundreds 5? Careful: the two places are not next to each other — count them.

Answer: _____

9. Estimate $6238 - 2905$ by rounding each number to the nearest hundred — that is, to the nearest 10^2 .

Write both rounded numbers, then the estimated difference. Look closely at the tens place of 2905 before you round it.

Answer: _____

10. Find $4500 \div 100$.

Dividing by a power of ten shifts the digits the opposite way from multiplying. Say how many places the digits move and in which direction, then give the quotient.

Answer: _____

11. A number is written in expanded form using powers of ten:

$$3 \times 10^4 + 7 \times 10^2 + 5$$

(a) What is the value of each term?

(b) Write the number in standard form. Two places have no term written for them — decide what digit belongs in each of those places and why.

Answer: _____

12. A warehouse counts 8120 stickers and packs them into boxes of 39.

Estimate $8120 \div 39$ by first rounding the divisor to the nearest ten, so that the division becomes a shift by a power of ten.

(a) What does 39 round to? (b) Estimate the quotient. (c) Is your estimate larger or smaller than the exact quotient? Explain in one sentence why rounding the divisor *up* does that.

Answer: _____